Here "1 project carries 5 marks" is equivalent to saying "2 projects carry 10 marks" and so on and hence the ratios, 1 : 5 = 2 : 10 = 3 : 15 = 4 : 20 are said to be in Proportion. Thus 1 : 5 is in proportion to 2 : 10, 3 : 15, 4 : 20 and so on. This is denoted by 1 : 5 : : 2 : 10 and it is read as '1 is to 5 as 2 is to 10' and so on.

Situation 2



Fig. 3.3(a)

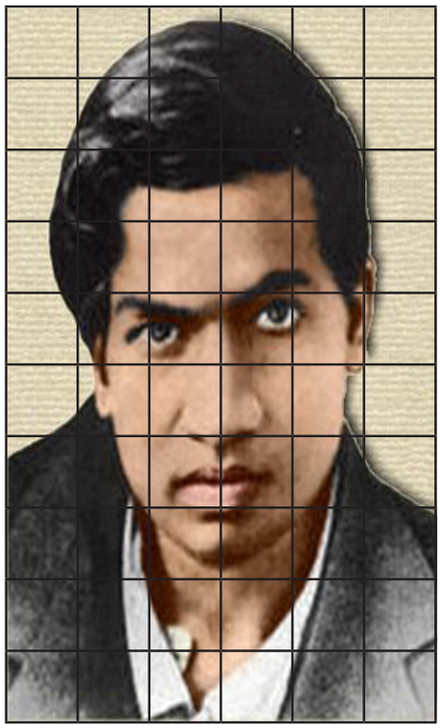


Fig. 3.3(b)

The size of the photograph of Srinivasa Ramanujan as shown in Figure 3.3 (a) is of length 5 grids and breadth 3 grids. Figure 3.3 (b) shows the enlarged size of the photograph of length 10 grids and breadth 6 grids. Here,

Photo grid length	:	Enlarged Photo grid length	=	5 : 10 (1 : 2)	and	Photo grid breadth	:	Enlarged Photo grid breadth	=	3 : 6 (1 : 2)

As the two ratios are equal, the given figures are in proportion. This is represented as 5 : 10 : : 3 : 6 or 5 : 10 = 3 : 6 and it is read as '5 is to 10 as 3 is to 6'

3.3.1 Proportionality law

If two ratios are in proportion ie., $a : b : : c : d$ then the product of the extremes is equal to the product of the means ad=bc. This is called the proportionality law. Here, a and d are the extremes and b and c are the means. Also, if two ratios are equal ie., $\frac{a}{b} = \frac{c}{d} \rightarrow ad = bc$ is called the cross product of proportions.

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